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Gate CS 2010,Qustion no.6

QUESTION

The minterm expansion of

$$F = PQ + QR' + PR'$$

(A) $m_2 + m_4 + m_6 + m_7$ (B) $m_0 + m_1 + m_4 + m_5$

(C) $m_0 + m_1 + m_6 + m_7$ (D) $m_2 + m_3 + m_6 + m_7$

QUESTION ANALYSIS

$$F = PQ + QR' + PR'$$

Where:

- P, Q, R are input variables
- R' is complement of R
- '+' represents OR operation
- Concatenation represents AND operation

Output will be displayed as:

- 0 when expression evaluates to 0
- 1 when expression evaluates to 1

TRUTH TABLE

P	Q	R	F
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

Minterm form:

$$F = \Sigma m(2, 4, 6, 7)$$

HARDWARE IMPLEMENTATION

The Boolean expression is implemented using:

- Arduino UNO (ATmega328P)
- 7447 BCD to 7-segment decoder
- Common Anode 7-segment display

Inputs are given through PORTD pins and output is sent through PORTB to 7447.

COMPONENTS REQUIRED

Sl.No	Component
1	Arduino UNO
2	7447 IC
3	Common Anode 7-Segment Display
4	220Ω Resistors (7 Nos)
5	Breadboard
6	Connecting Wires

PIN CONNECTIONS

Input Connections

Variable	Arduino Pin	AVR Port
P	D2	PD2
Q	D3	PD3
R	D4	PD4

Output Connections (To 7447)

7447 Pin	Arduino Pin	AVR Port
A	D8	PB0
B	D9	PB1
C	D10	PB2
D	D11	PB3

LOGIC DESCRIPTION

The program reads inputs P, Q and R from PORTD.

- R is complemented to generate R'
- AND operations compute PQ, QR', and PR'
- OR operation combines all terms
- Result is sent as BCD to 7447

If F = 0, display shows 0. If F = 1, display shows 1.

CODE UPLOADING STEPS

1. Create a folder
2. Write the code in filename.asm
3. Run the command “avra filename.asm”. It will compile the code and creates hex file
4. Copy the .hex file to ArduinoDriod folder
5. Connect the Arduino UNO to mobile with OTG cable
6. Upload the hex file using “upload precompiled” option
7. Observe the output and verify the expression

EXPERIMENTAL TRUTH TABLE

P	Q	R	Observed F
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

CIRCUIT CONNECTION

CONCLUSION

The Boolean expression $F = PQ + QR' + PR'$ was successfully implemented using AVR Assembly language. The output was verified using 7447 decoder and Common Anode 7-segment display.